

Catch every strip defect where it is born, not where it is shipped

AI surface inspection that finds, grades and routes every stainless strip defect to the process that caused it.

0.764

accuracy on unseen steel
(published range 0.70-0.80)

93.7% → 32.5%

clean steel wrongly flagged,
before vs after our fix

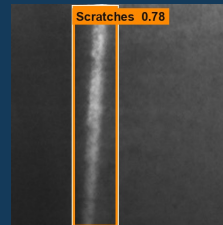
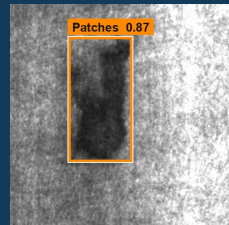
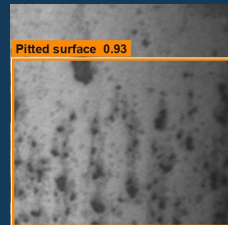
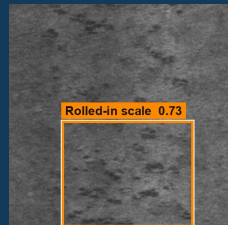
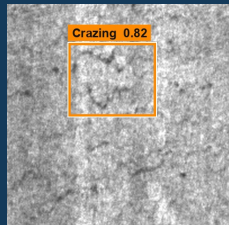
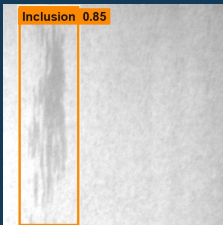
₹2.5-4.2 cr

per line, vs ₹10-29 cr for a
commercial system

~2.2 years

payback on one line at a
1% downgrade rate

REAL DETECTIONS BY OUR MODEL, ON STEEL IMAGES IT NEVER SAW IN TRAINING



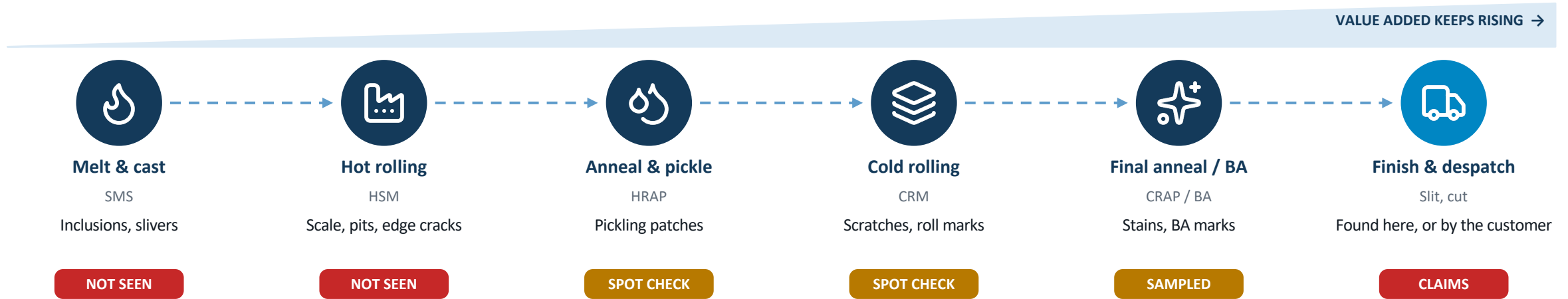
TRY THE LIVE DEMO

surface-vision.github.io

Runs in your browser. Nothing leaves the page.

A defect born at the caster is paid for at every later stage, but today it is found only at the end

Where stainless surface defects are born, and where they are seen today



Why it hurts more on stainless



300 m/min

Too fast for the human eye to see the whole surface



₹1.26-1.65 lakh

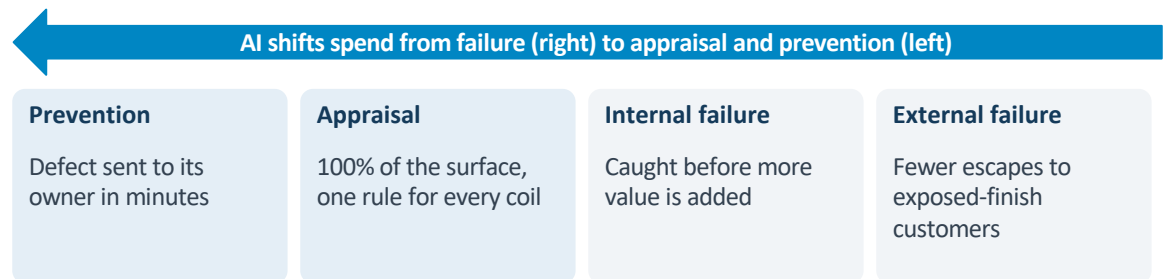
of nickel in every tonne of 304, given away on a downgrade



2.57 Mt

JSL FY26 sales: each 0.1% downgraded is ~2,570 t

Cost of quality (PAF): where AI inspection moves the money



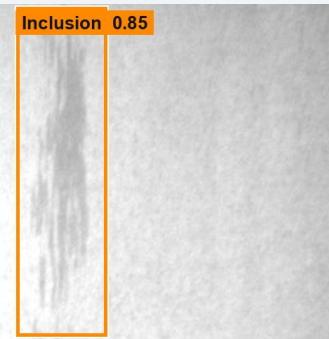
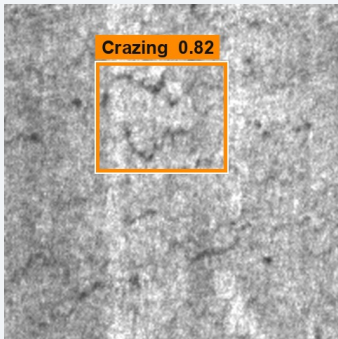
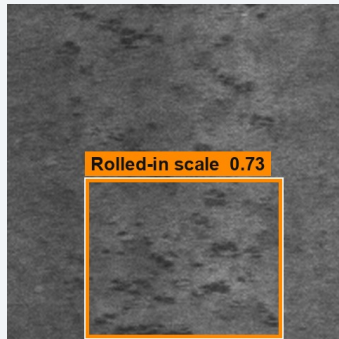
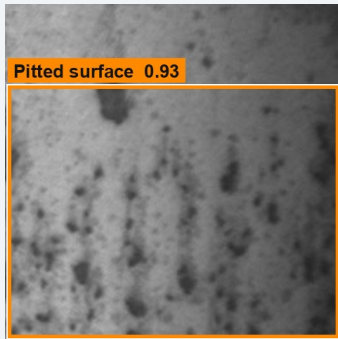
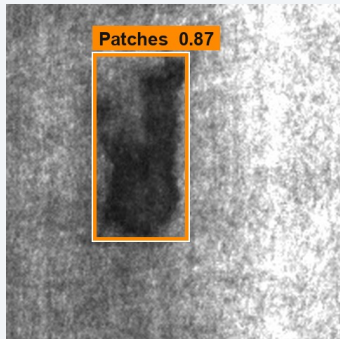
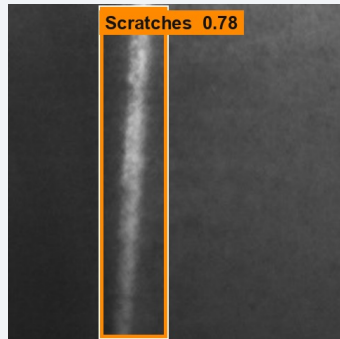
Cost of poor quality averages ~15% of sales in manufacturing (range 5-35%, IISE).

Sources: AMETEK Surface Vision / Ternium case study (2020); IISE, Measuring the Cost of Quality; ASTM A240; LME nickel USD 16,402/t at INR 95.8/USD (22 Sep 2026); JSL FY26 results.

The problem is not that defects exist. They are found late, by sampling, and nobody is told which process caused them.

On stainless the surface is the product, so every defect type must reach the process that caused it

Six defect families, six different owners (real detections by our model)

 Inclusion 0.85	 Cracking 0.82	 Rolled-in scale 0.73	 Pitted surface 0.93	 Patches 0.87	 Scratches 0.78
Born: Caster Owner: SMS Fix: Tundish level, mould flux	Born: Reheat, hot rolling Owner: HSM Fix: Reheat and coiling temperature	Born: Hot rolling Owner: HSM Fix: Descaler pressure, nozzles	Born: Reheat, pickling Owner: HSM / pickling Fix: Furnace atmosphere, over-pickling	Born: Anneal & pickle Owner: Pickling line Fix: Acid strength, line speed	Born: Cold rolling, handling Owner: CRM / logistics Fix: Guides, rolls, coil handling

Three facts that change the brief

-  The finish cannot be repaired. 2B/BA damage cannot be polished out (ASSDA).
-  Surface-critical tonnes are growing. Cold rolling 2.05 → 2.67 MTPA by FY28.
-  End markets see the surface. Auto, rail, metros, white goods (Q1 FY27).

Grade-by-defect risk: which defects to train for first

Grade	Slivers	Scale	Scratches	Edge cracks	Ridging
200 series (Cr-Mn)	M	M	H	M	L
300 series (304, 316)	H	M	H	L	L
400 series (409, 430)	H	M	M	M	H
Duplex (2205)	M	H	M	H	L

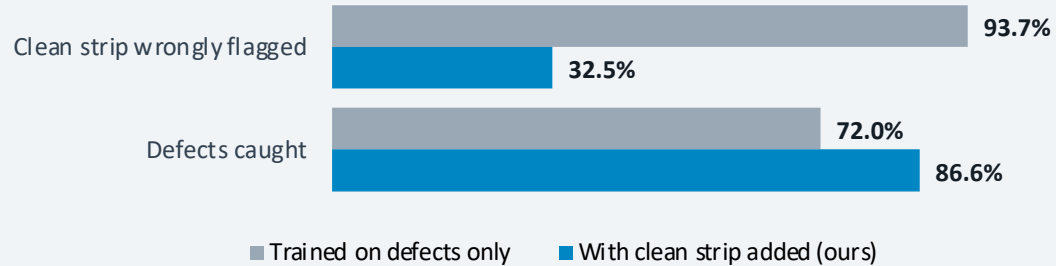
OUTPUT
300 series is the pilot grade:
highest exposed-finish risk and
the most nickel per tonne.

Images: held-out test frames, boxes drawn by our model. Sources: ASSDA; JSL Q1 FY27 results; Leão et al. 2021; Steel in Translation 2023. Owners and grade risks: team hypotheses, to validate with JSL.

Classify, don't just alarm: a typed defect becomes a work order for a named owner. The pilot belongs on 300-series cold-rolled strip.

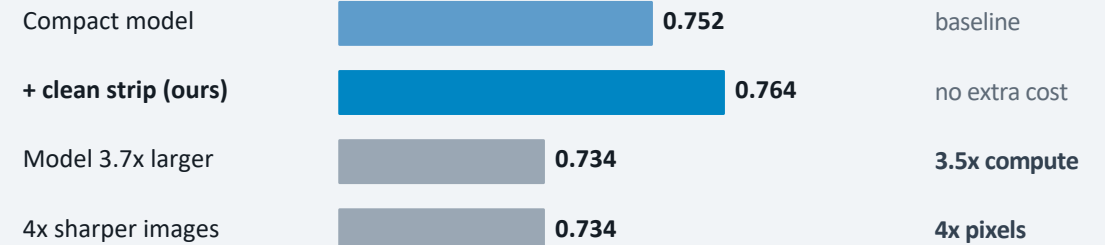
Our tests produced four findings, and each one changed the design

1 A model that never saw clean steel cries wolf



So we: train on JSL's own clean and defective strip before any alarm is trusted.

2 A bigger model or sharper images did not help



Accuracy on held-out images; bars start at 0.70. Compact vs large is within noise.

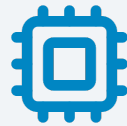
So we: keep the model compact and spend the budget on camera coverage.

3 Line speed, not the model, sets the hardware bill



~18 processors
see every 0.2 mm of a 1.28 m strip at 250 m/min

vs



1 processor
if images are coarsened to 1.28 mm, but fine scratches vanish

So we: fix the image resolution with JSL first, then size the hardware.

4 Operators act only on alarms they can trust



68% lower

confidence error after calibration. A "60% sure" alarm used to be right 84% of the time; now it is within ~5 points



0 of 3,200

real steel images wrongly rejected by our "is this steel?" check, which stops the system judging anything else

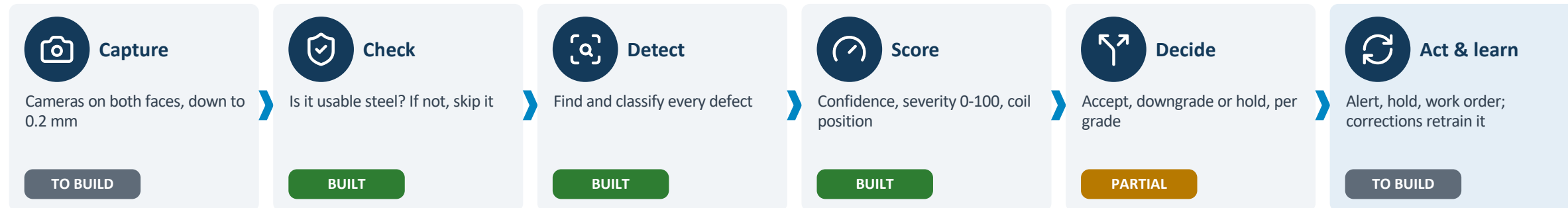
So we: every alarm carries a trustworthy confidence and a steel check.

All numbers measured by us on three public steel-surface datasets, on images the model never saw in training.

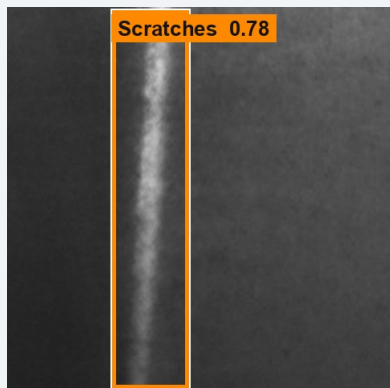
The model is the cheap part. JSL's own strip, the image resolution and trustworthy confidence decide whether the system works.

Detect on the strip, decide at the coil, route to the owner, and learn from every correction

How it works: six stages from camera to corrective action



Running today



Upload a strip image, get:

- box on the defect
- defect type
- confidence
- severity score

surface-vision.github.io runs in the browser

What is new beyond a standard detector

- Trained on clean strip too**
false alarms 93.7% → 32.5%
- Root-cause routing**
every alarm becomes a work order
- Unseen-defect check**
flags patterns never labelled (proposed)
- Learns from unlabelled coils**
JSL's image archive becomes an asset

Build vs buy

	Cost per line	Proven at speed	Tuned to JSL grades	JSL owns model
Commercial USD 1-3M (unverified)	●	●	●	●
Fully in-house ₹2-3 cr (est.)	●	●	●	●
Hybrid (ours) ₹2.5-4.2 cr (est.)	●	●	●	●

● strong ● partial ● weak

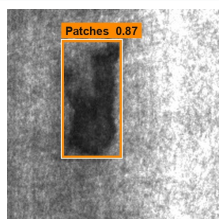
Output: hybrid is never weak.

BUILT = working in our prototype today, live at surface-vision.github.io. Unlabelled-learning result: J. Mater. Inf. 2025. Commercial price range: industry figure, unverified.

Buy proven cameras, own the model: vendor-grade capture with a detector trained on JSL's grades and wired to JSL's process owners.

It already works on public steel data; the pilot's job is to prove it on JSL's own strip

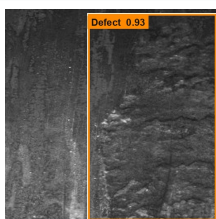
Tested on three public steel datasets



NEU-DET

1,800 images, 6 defect types

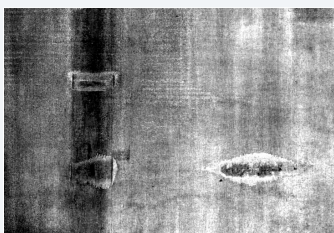
University benchmark, hot-rolled strip



Severstal

12,568 real mill frames

Production strip from a steel mill



GC10-DET

2,294 frames, 10 types

Full-width line-scan camera frames (roll marks shown)

How well it finds each defect type (accuracy, held-out images)



published range

Headline results



176 of 180

test images: the right defect type named



27 ms

per image, running in an ordinary laptop browser



~10,100

images across three datasets, for training and testing

Known weak spots, and the fix for each

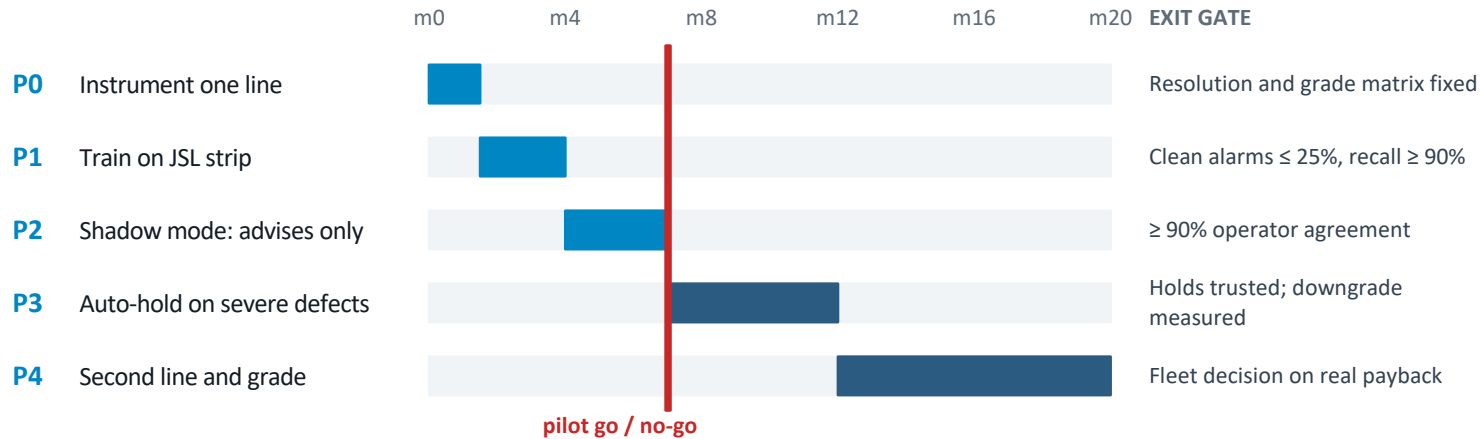
Weak spot	Today (0-1)	Fix in the pilot
Low-contrast texture: crazing, rolled-in scale	0.59-0.60	Angled lighting; JSL labels in P1
Real mill frames are harder than the benchmark	0.46 vs 0.76	Train on JSL's own strip in P1
Roll marks and edge cracks: few examples	0.20	Edge cameras; targeted labelling

Datasets: NEU-DET (Northeastern Univ.); Severstal steel defects (Kaggle); GC10-DET (Lv et al. 2020, CC BY 4.0). All scores on held-out images the model never saw in training.

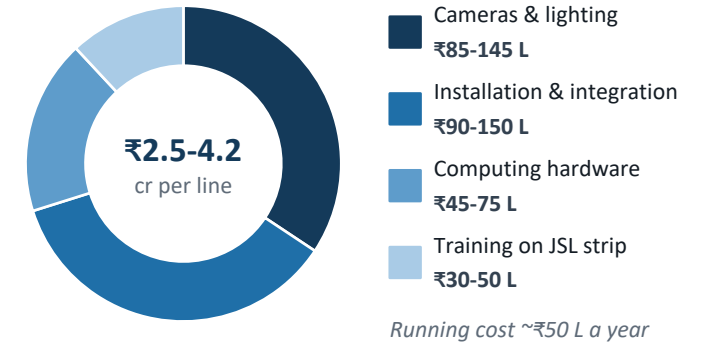
Strong on four of six defect families today. Every weak spot is known, and each has a fix in the pilot plan.

Seven months to shadow mode on one line, and nothing touches the mill until trust is earned

Stage-gate roadmap (months)



Cost per line, estimated



Top risks, and when each is retired

Risk	Mitigation	Retired
Model does not transfer to JSL strip	Train on JSL clean and defect images	P1
Low-contrast defects missed	Angled lighting, sharper images	P0-P1
Rare defects under-learned	Edge cameras, targeted labelling	P1-P3
Operators ignore alarms	Trusted scores, shadow mode first	P2
Heat, dust, acid fumes	Sealed, cooled enclosures	P0

Pilot KPIs: how JSL will know it works

 $\geq 90\%$ defective strip flagged
measured 92.2% **$\leq 25\%$** clean strip wrongly flagged
measured 32.5% **≤ 5 pts**confidence vs reality gap
measured 4.6 pts **< 0.2 s**defect to operator screen
design target **$\geq 90\%$** operator agreement, shadow
measured in P2 **< 8 h**retrain for a new grade
measured 0.6 h

Costs are team estimates for a 1.3-1.6 m line inspected on both faces, at INR 95.8/USD, to be replaced by vendor quotes in P0. KPI targets to be agreed with JSL Quality in P0.

Commit ₹2.5-4.2 cr and one line for seven months. The month-7 go/no-go is decided on JSL's own alarm and recall numbers.

At a conservative 1% downgrade rate one line pays back in ~2.2 years; the fleet protects ~₹13 cr a year

Loss pool per line, ₹ cr a year

$0.4 \text{ Mt} \times \text{downgrade rate} \times \text{price discount} \times ₹1,67,407/\text{t}$

Downgraded	10% off	15% off	20% off
0.5% of volume	3.3	5.0	6.7
1.0% of volume	6.7	10.0	13.4
2.0% of volume	13.4	20.1	26.8

Base case (dark): ₹6.7 cr pool. Recover 30% = ₹2.0 cr/yr, ₹1.5 cr net of run cost. Fleet (2.57 Mt): ₹43 cr pool, ₹12.9 cr/yr recovered.

Payback per line, base case

Hybrid (ours)

2.2 yr

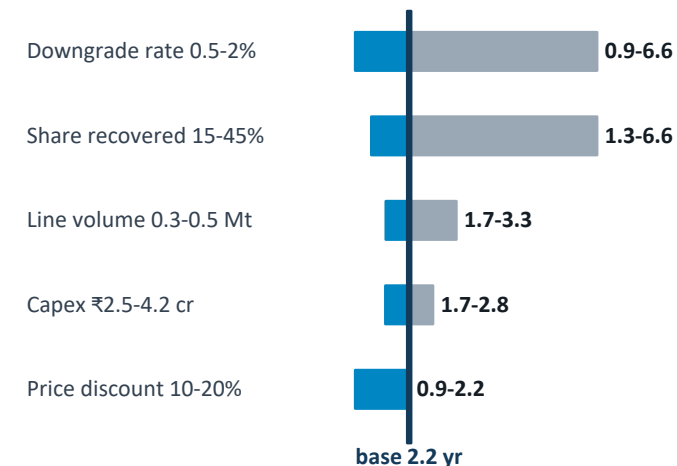
Commercial system

12.7 yr

+₹2.1 cr 5-year NPV per line at 12%

≥ 0.11% of line revenue lost to downgrade is enough for a 2-year payback

What moves payback (years)



Impact beyond the P&L



Coverage

100% of both faces, every coil



Customers

Defect map per coil, fewer claims



Process

Defects tied to heat and coil IDs



Safety

Severe-defect holds prevent strip breaks



Sustainability

Less remelt, acid and nickel waste



People

Inspectors move to decisions

WHAT WE ASK OF JSL

- One 300-series cold-rolled line, 7 months
- ₹2.5-4.2 cr pilot budget
- Downgrade rate by line and grade
- A metallurgist at 20% for root causes

In return: measured downgrade rate, payback and a go / no-go at month 7.

Inputs: JSL FY26 (INR 1,67,407/t; 25,65,902 t). Base: 0.4 Mt line, 30% of loss recovered, ₹3.35 cr capex, ₹0.5 cr/yr run cost, 12% rate. Commercial: USD 2M at INR 95.8.

Recommendation: approve a 7-month pilot on one 300-series cold-rolled line. JSL's own downgrade rate decides fleet rollout.